

Task 12: Advanced Constraints & Data Validation

Tools:

- MySQL Workbench
- Alternatives: PostgreSQL, BigQuery Sandbox, SQLite

Hints / Mini Guide:

1. Create tables using CHECK, DEFAULT, and UNIQUE constraints.
2. Validate numeric ranges using CHECK constraints.
3. Apply DEFAULT values for timestamps.
4. Test constraint violations intentionally.
5. Combine multiple constraints on a single column.
6. Understand constraint enforcement order.
7. Compare application-level vs database-level validation.
8. Document constraint design decisions.

Deliverables:

- SQL schema demonstrating constraint usage

Final Outcome:

- Intern ensures strong data quality and validation.

Interview Questions Related To Above Task:

- What is CHECK constraint?
- Why use DEFAULT?
- Can constraints be removed later?
- UNIQUE vs PRIMARY KEY?
- Database vs application validation?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📤 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

